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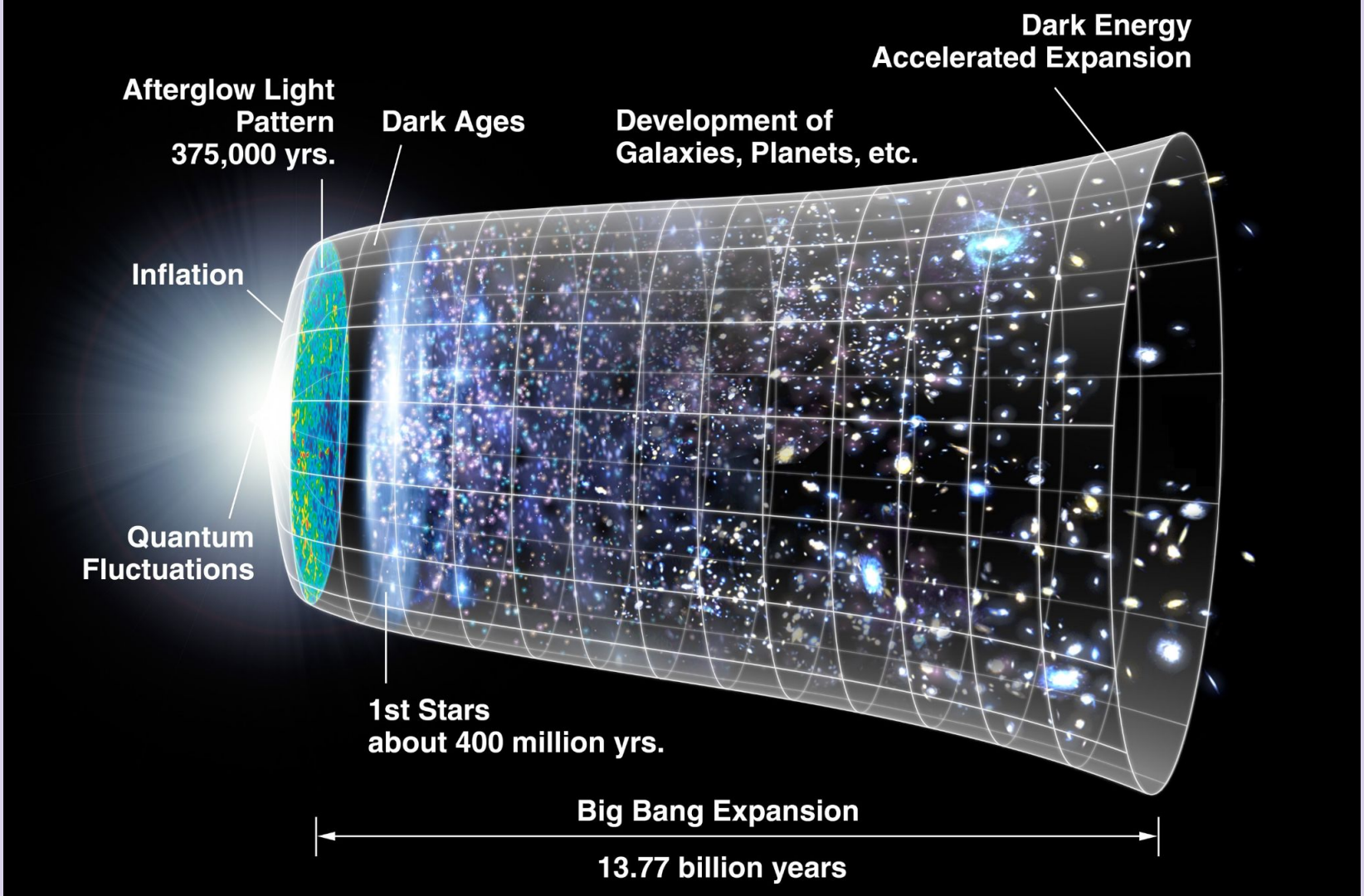
Go to:

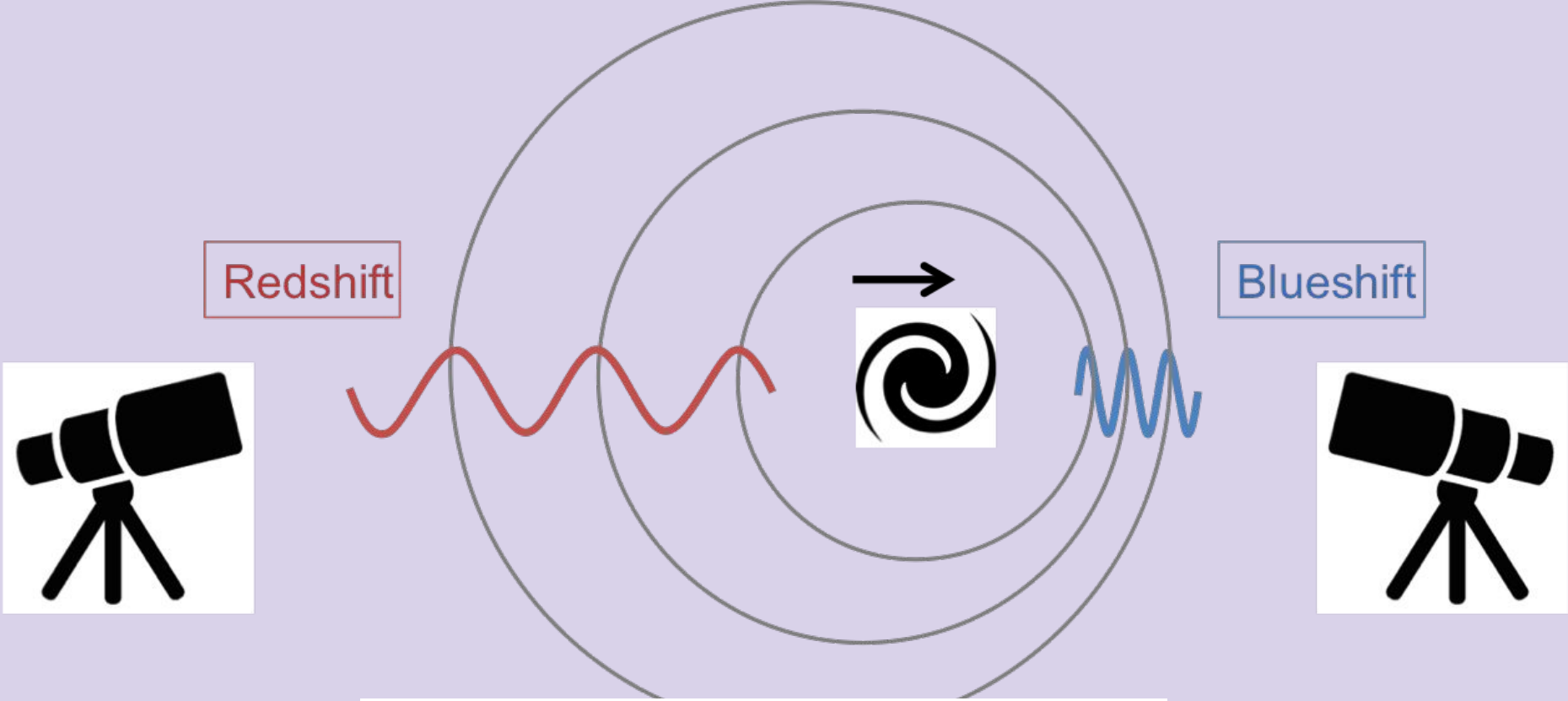
https://github.com/ianpajohnsonSBU/WISE_computing_heavens

And download today’s files

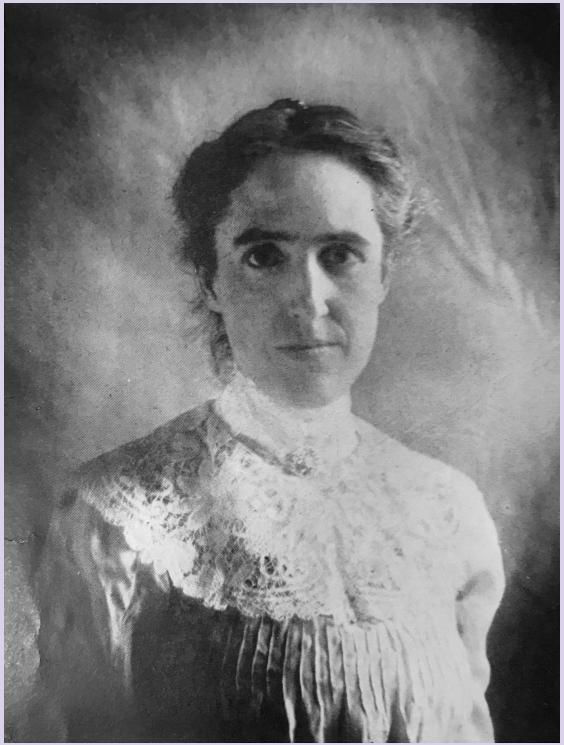
Cosmology

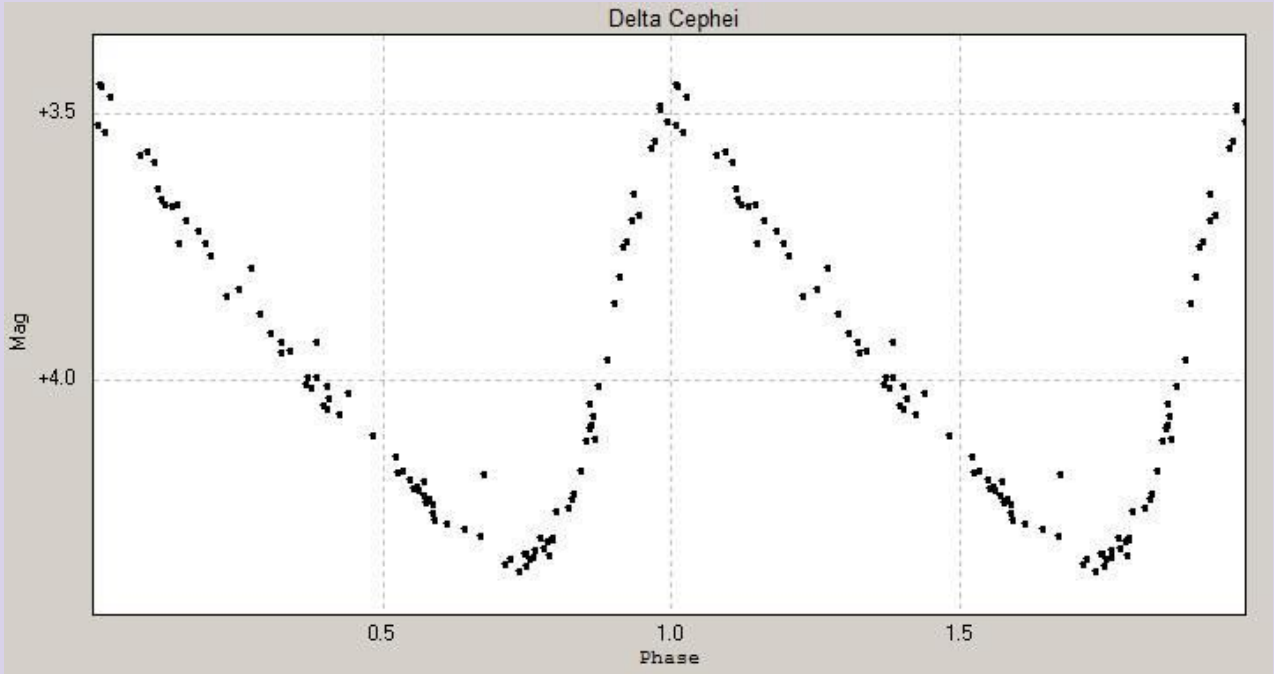
Ian Johnson
For SBU HS WISE


Dark Energy
Accelerated Expansion
Afterglow Light
Pattern
375,000 yrs.
Dark Ages
Development of
Galaxies, Planets, etc.
Inflation
Quantum
Fluctuations
1st Stars
about 400 million yrs.
Big Bang Expansion
13.77 billion years

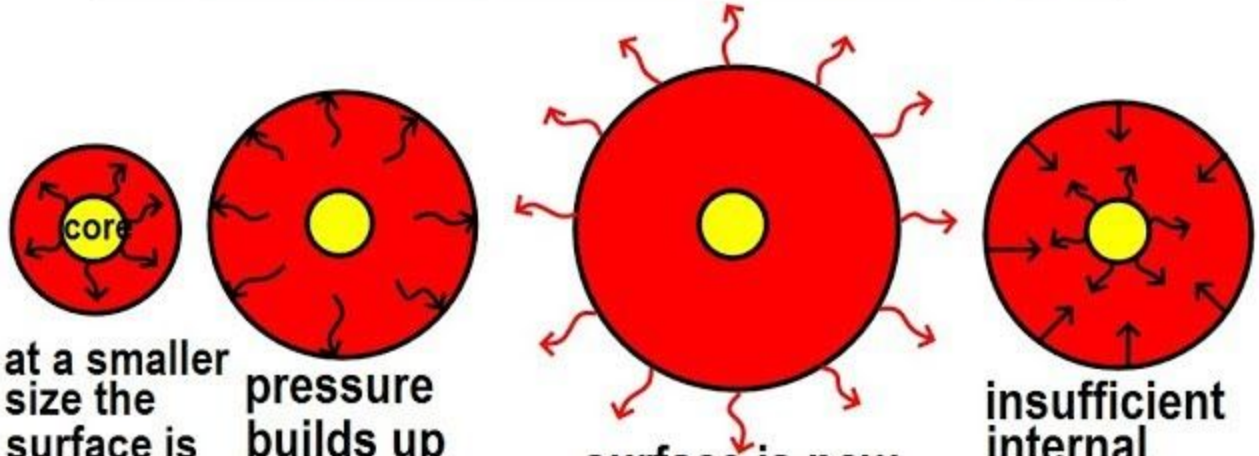


$$z = \frac{\Delta\lambda}{\lambda_{rest}} = \frac{\lambda_{obs} - \lambda_{rest}}{\lambda_{rest}} = \frac{v}{c}$$




Delta Cephei
+3.5
+4.0
Mag
0.5
1.0
1.5
Phase

Cepheid Variables: Why Do they Vary?

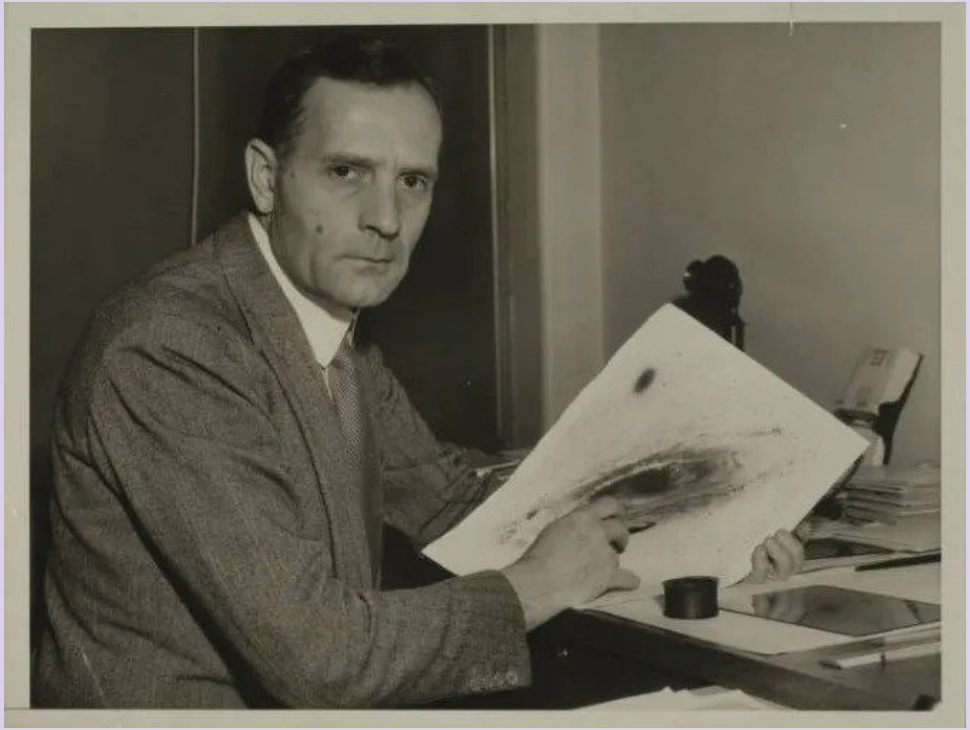


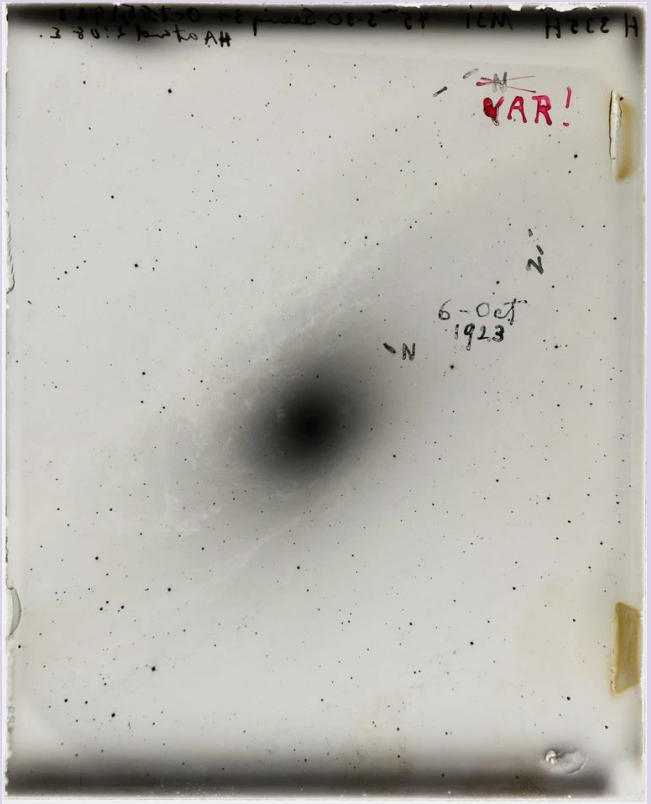
at a smaller size the surface is more opague to the...

pressure builds up expanding the size of the star

surface is now cooling from the increased radiation to space

insufficient internal pressure causes the star to contract

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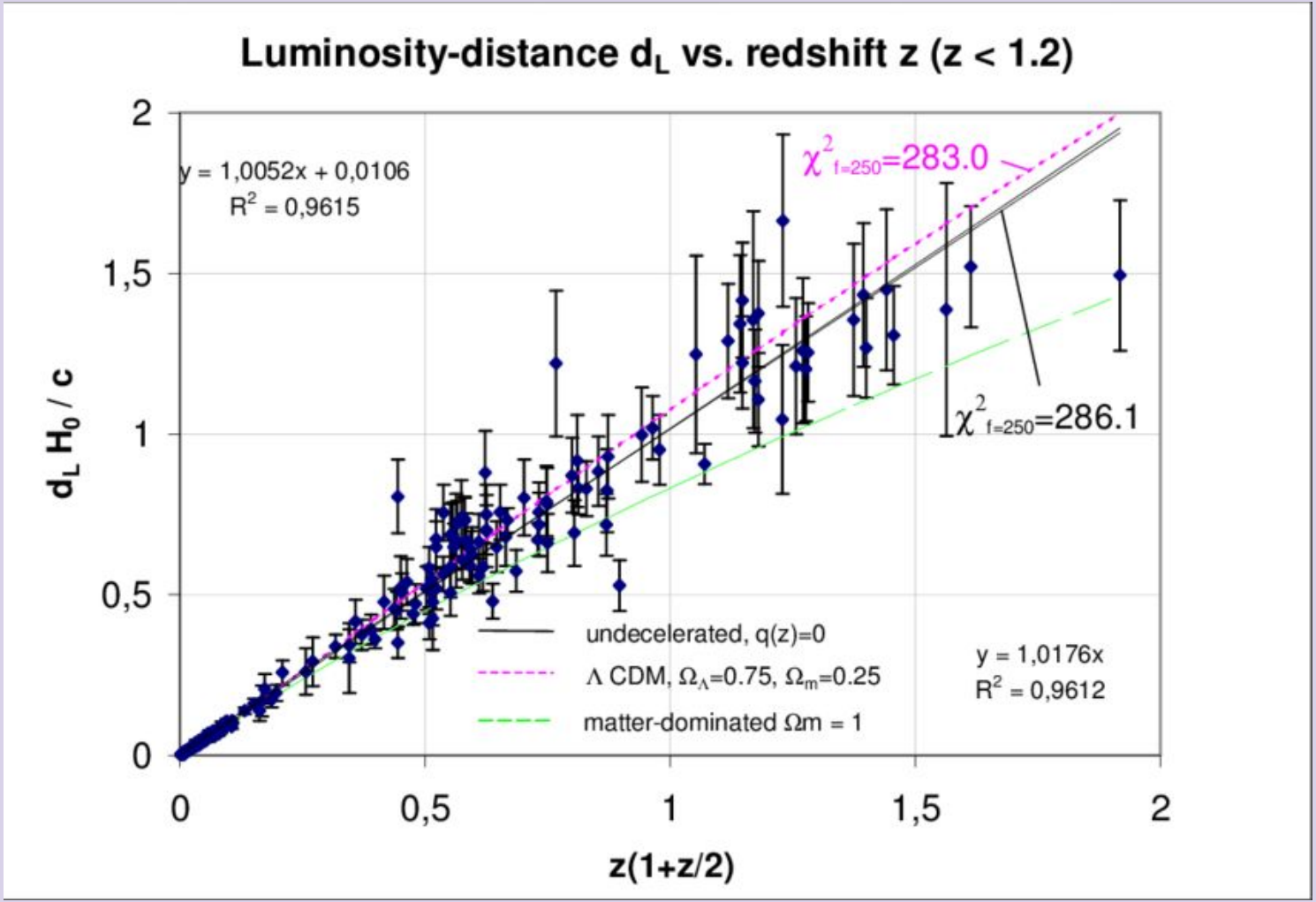
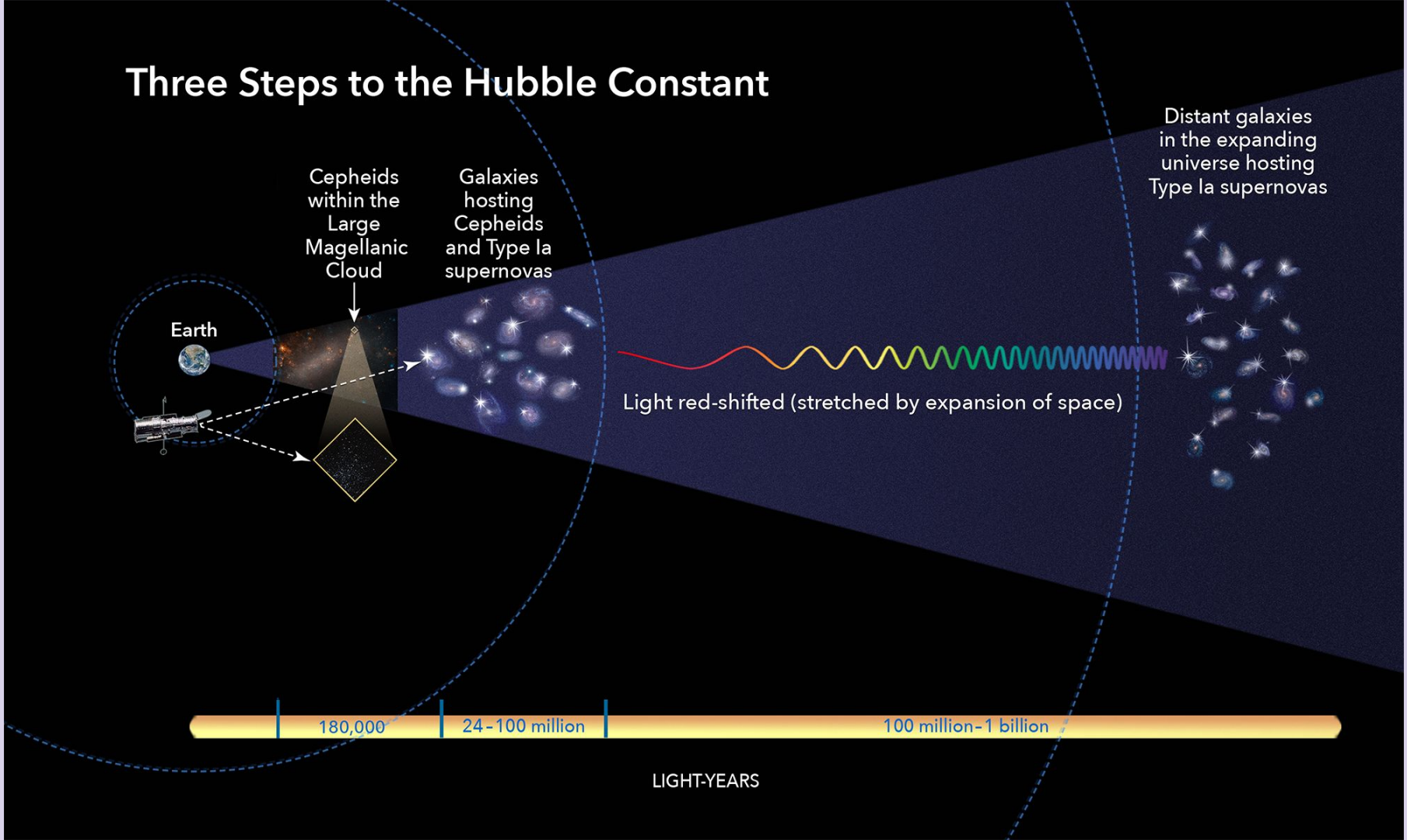

Luminosity-distance d_L vs. redshift z (z < 1.2)
y = 1,0052x + 0,0106
R^2 = 0,9615
$\chi^2_{f=250}$=283.0
$\chi^2_{f=250}$=286.1
undecelerated, q(z)=0
Λ CDM, Ω_Λ=0.75, Ω_m=0.25
matter-dominated Ωm = 1
y = 1,0176x
R^2 = 0,9612
$d_L H_0 / c$
z(1+z/2)
0
0,5
1
1,5
2


Three Steps to the Hubble Constant
Cepheids within the Large Magellanic Cloud
Galaxies hosting Cepheids and Type Ia supernovas
Distant galaxies in the expanding universe hosting Type Ia supernovas
Earth
Light red-shifted (stretched by expansion of space)
180,000
24-100 million
100 million-1 billion
LIGHT-YEARS